



JURONG JUNIOR COLLEGE
2018 JC 2 PRELIMINARY EXAMINATION

CANDIDATE
NAME

CLASS

18S24

EXAM INDEX

CHEMISTRY

8873/02

Higher 1

Paper 2 Structured Questions

29 August 2018

2 hours

Candidates answer on the Question Paper.

Additional Materials: Data Booklet

READ THESE INSTRUCTIONS FIRST

Write your name, class and exam index number on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use an HB pencil for any diagrams, graphs.

Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in **Section A (60 marks)** in the space provided on the Question paper.

Answer **one** question in **Section B (20 marks)** in the space provided on the Question paper.

The use of an approved scientific calculator is expected, where appropriate.

A Data Booklet is provided.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
1	
2	
3	
4	
5	
6	
Section B	
Lack 3sf in final ans	-1 / NA
Missing/wrong units in final ans	-1 / NA
Total	

This document consists of **20** printed pages and **1** blank page.

Section A

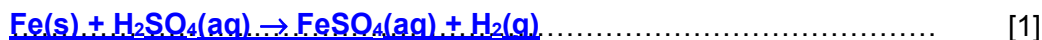
Answer **all** the questions in this section, in the space provided.

For
Examiner's
Use

- 1 An experiment was carried out to determine the percentage of iron in a sample of iron wire.

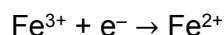
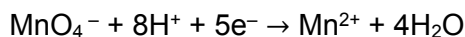
- (a) A 3.35 g piece of wire was reacted with dilute sulfuric acid, in the absence of air, so that all of the iron atoms were converted to iron(II) ions. The resulting solution was made up to 250 cm³.

- (i) Write a balanced equation for the reaction between the iron in the wire and the sulfuric acid.

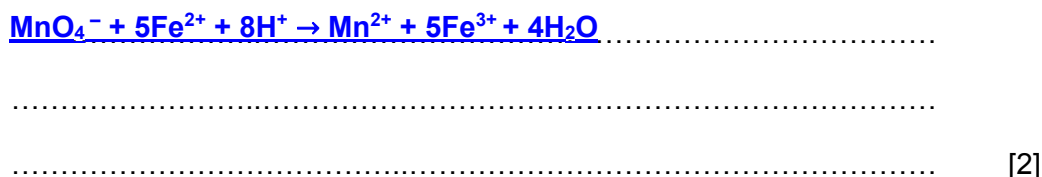


A 25.0 cm³ sample of this solution was acidified and titrated with 0.0250 mol dm⁻³ potassium manganate(VII). 32.0 cm³ of the potassium manganate(VII) solution was required for complete reaction.

The relevant half-equations for the reaction are shown below.



- (ii) Use these two half-equations to construct an ionic equation for the reaction between manganate(VII) ions and iron(II) ions in acid solution.



- (iii) Calculate the amount, in moles, of manganate(VII) ions used in the titration.

Amount of MnO_4^- used = $32.0/1000 \times 0.025 = \underline{8.00 \times 10^{-4} \text{ mol}}$

[1]

- (iv) Calculate the total amount, in moles, of iron(II) ions in 250 cm³ of the solution.

Amount of Fe^{2+} in 25.0 cm³ = $\underline{8.00 \times 10^{-4} \times 5} = 4.00 \times 10^{-3} \text{ mol}$

Amount of Fe^{2+} in 250 cm³ = $4.00 \times 10^{-3} \times 250/25.0 = \underline{4.00 \times 10^{-2} \text{ mol}}$

[2]

- 1 (a) (v) Hence, calculate the percentage by mass of iron in the 3.35 g piece of wire.

$$\text{Mass of Fe in wire} = 4.00 \times 10^{-2} \times 55.8 = 2.23 \text{ g}$$

$$\% \text{ by mass of Fe in wire} = 2.23/3.35 \times 100 \% = 66.6 \%$$

[2]

- (b) The electronegativity values of some elements are shown in the table below.

element	electronegativity
aluminium	1.5
chlorine	3.0
iron	1.8

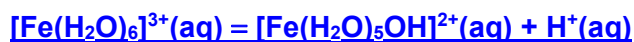
- (i) Use the data to suggest the nature of the bonding in iron(III) chloride. Explain your answer.

Covalent. Smaller difference in electronegativity between Fe and Cl compared to between Al and Cl.

[2]

- (ii) When a piece of universal indicator paper is dipped into a solution of iron(III) chloride, the indicator paper turns red.

Suggest an equation to account for the colour change in the indicator paper.



[1]

[Total: 11]

- 2 (a) State the difference between a nanoparticle and a nanomaterial in terms of size.

Nanoparticles ALL dimensions 1–100nm/on the nanoscale
AND nanomaterial (at least) one dimension 1–100nm/on the nanoscale

[1]

- (b) Many modern sunscreens contain nano-sized particles of titanium dioxide. This substance does **not** absorb ultraviolet radiation.

Suggest how these nano-particles are able to protect skin from ultraviolet radiation.

The particles are similar in size to the wavelength of uv light and deflect
the harmful radiation

[1]

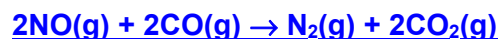
- (c) The engines of modern motor cars have exhaust systems which are fitted with catalytic converters in order to reduce atmospheric pollution from substances such as nitrogen monoxide, NO. These substances are converted to less harmful gases on the catalyst surface before they are released into the atmosphere.

- (i) One of the active materials present in the catalytic converter is platinum nanoparticles. Explain how the use of nanoparticles helps to increase the rate of gaseous reactions in the catalytic converter.

Pt nanoparticles are used due to the high surface area to volume ratio
Due to the large surface area, more platinum atoms are exposed to the
reactants. Hence, the catalyst provides an alternative reaction pathway of lower
activation energy.
the greater the frequency of effective collisions between the reactants and
catalyst, the higher the reaction rate.

[2]

- (ii) Give an equation for a reaction involved in the removal of nitrogen monoxide from the exhaust gases of the car, in the catalytic converter.



[1]

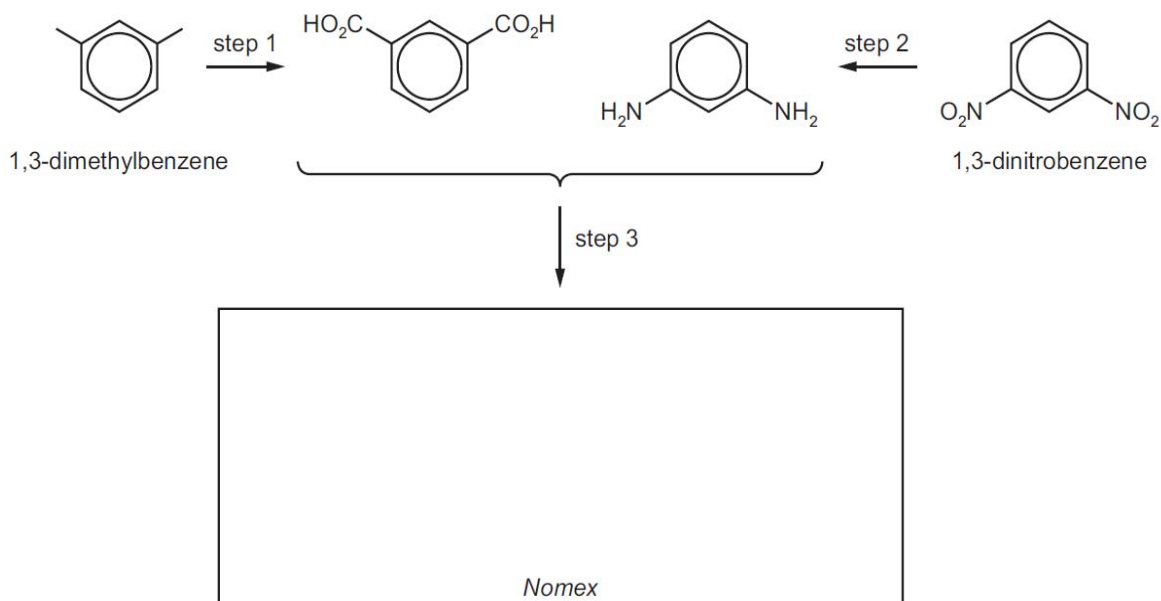
- (iii) State one way in which nitrogen monoxide acts as air pollutant.

Formation of acid rain / causes smog

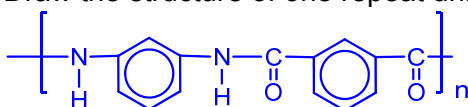
[1]

[Total: 6]

- 3 1,3-dimethylbenzene is a starting material for the synthesis of the polymer *Nomex*, used in fireproof protective clothing worn by firefighters, military pilots and racing car drivers. The polymer is made from 1,3-dimethylbenzene and 1,3-dinitrobenzene by the following route.



- (a) (i) Draw the structure of one repeat unit of *Nomex* in the box above.



[1]

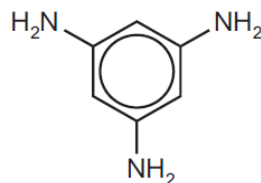
- (ii) State the type of reaction that occurs in step 3.

condensation polymerisation..... [1]

- (iii) Suggest the by-product formed in step 3.

H₂O / water [1]

- (iv) Suggest how and why the properties of the polymer might change if some of the diamine monomer were replaced with 1,3,5-triaminobenzene.



1,3,5-triaminobenzene

Harder/more dense/stronger/higher melting point/tougher/more rigid due to cross-linking/more hydrogen bonding between the chains

[1]

3 (b) Similar to *Nomex*, Nylon is also a polyamide.

(i) State one use, with reason, of nylon.

rope as it is strong / fabric as it is lightweight

[1]

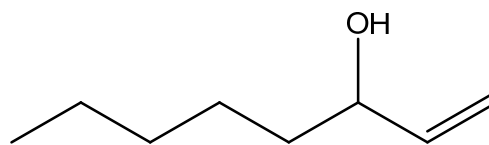
(ii) Nylon is non-biodegradable. What does this mean?

Cannot be broken down / changed to a harmless natural state by the action of bacteria / living organism

[1]

[Total: 6]

4 1-octen-3-ol has the following structure.



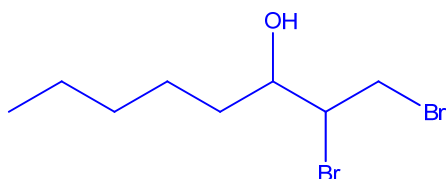
This chemical is found in human sweat and breath, and is thought to attract biting insects such as mosquitoes. It is also found in mushrooms, and sometimes called 'mushroom alcohol'.

(a) (i) State the type of reaction and what you would observe when 1-octen-3-ol reacts with a solution of bromine.

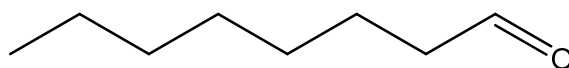
Electrophilic addition

Orange-red bromine decolourises.

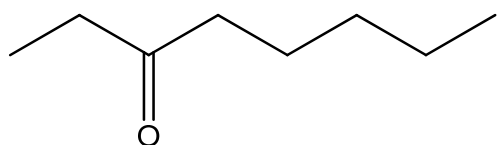
(ii) Draw the structure of the organic product formed when 1-octen-3-ol reacts with bromine in tetrachloroethane.



(b) Other examples of organic compounds containing eight carbon atoms are octanal and octan-3-one.



octanal



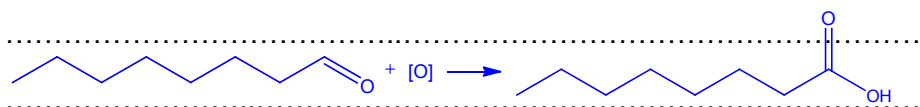
octan-3-one

(i) Octanal and octan-3-one can both be reduced using sodium borohydride, NaBH_4 .

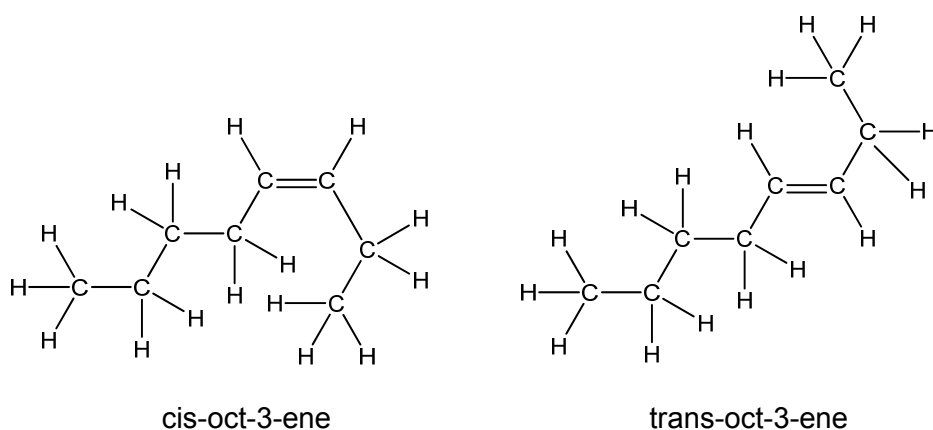
Draw the structural formula of the organic product obtained from the reduction of octanal and octan-3-one.

Organic compound	Organic product obtained
Octanal	
Octan-3-one	

4 (b) (ii) Octanal and octan-3-one were heated separately with acidified potassium dichromate(VI). Write the structural formula of the organic product obtained from the oxidation of octanal.

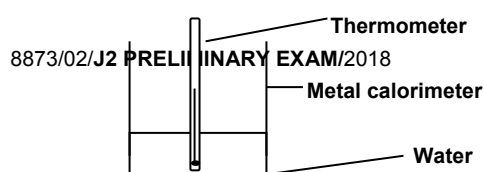


- (c) Oct-3-ene, C_8H_{16} , has two isomers with the same structural formula. Draw the **displayed** formulae of the two isomers and label them clearly.



5 Biofuels such as ethanol, methanol and methane have been investigated as possible alternative fuels for motor vehicles.

- (a) The enthalpy change of combustion of ethanol can be determined in the laboratory using the apparatus shown in Figure 1. The heat produced by the burning fuel warms a known mass of water.



The following results were obtained.

Mass of ethanol burnt ($M_r = 46.0$)	0.10 g
Initial temperature recorded	25.0 °C
Maximum temperature recorded	x °C
Enthalpy change of combustion of ethanol calculated	-840 kJ mol^{-1}

Assume the total heat capacity of the calorimeter and its contents is 120 J K^{-1} , calculate the maximum

Amount of ethanol burnt in the experiment

$$= \frac{0.10}{46.0} = 0.00217 \text{ mol}$$

Heat evolved by combustion of 0.10 g of ethanol

= Heat used to raise temp. of water

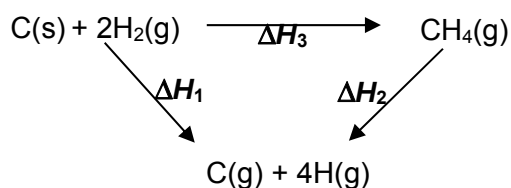
$$= 0.00217 \times 840 = 1.82 \text{ kJ} = 1820 \text{ J}$$

$$= 120 \times \Delta T$$

$$\Delta T = 15.2 \text{ K} = 15.2 \text{ °C}$$

$$\text{Final max temp} = 25.0 + 15.2 = 40.2 \text{ °C}$$

- 5 (b) Consider the energy cycle below involving methane and the enthalpy changes, ΔH_1 , ΔH_2 and ΔH_3 .



- (i) State Hess's Law.

Hess' Law states that the enthalpy change in a chemical reaction is constant and is independent of the reaction pathway between the initial and final states.

.....

- (ii) Using the data given below, calculate ΔH_1 .



$$\Delta H_1 = +715 + 2(+436) = \underline{+1587 \text{ kJ mol}^{-1}}$$

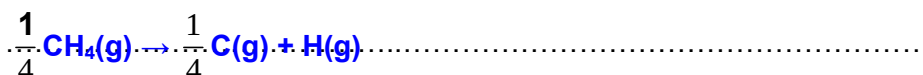
- (iii) What enthalpy change does ΔH_3 represent?

Standard enthalpy change of formation of methane

- (iii) Using your answer to (b)(ii) and $\Delta H_3 = -75 \text{ kJ mol}^{-1}$, calculate ΔH_2 .

$$\begin{aligned} \Delta H_2 &= \Delta H_1 - \Delta H_3 \\ &= \underline{+1587 - (-75)} \\ &= \underline{+1662 \text{ kJ mol}^{-1}} \end{aligned}$$

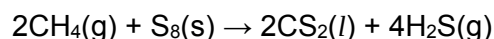
- (iv) Write an equation to represent the average C-H bond energy in methane, CH_4 .



- (v) Hence, calculate the average C-H bond energy in methane.

$$\frac{1}{4} \square \times (+1662) = \underline{+415.5 \text{ kJ mol}^{-1}}$$

- 5 (c) Methane and sulfur react to produce carbon disulfide (CS_2), a liquid often used in the production of cellulosic acetone.



- (i) Write an equation to represent the standard enthalpy change of combustion of liquid carbon disulfide.



The tables below contain values of standard enthalpy change of combustion, ΔH_c° , and standard enthalpy of formation, ΔH_f° , of various substances.

Table (I)

	C(graphite)	S(s)
$\Delta H_c / \text{kJ mol}^{-1}$	-394	-297

Table (II)

	CS ₂ (l)	CO(g)
$\Delta H_f / \text{kJ mol}^{-1}$	+88	-111

Use of relevant data given in **Tables (I) and (II)** is required to solve **c(ii)** and **c(iii)** of this question.

- (ii) Calculate the standard enthalpy change of combustion of CS₂(l).

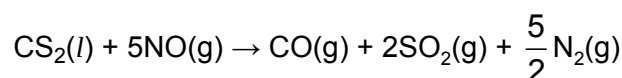
$$\Delta H_r = \Sigma m \Delta H_c (\text{reactants}) - \Sigma n \Delta H_c (\text{products})$$

$$\Delta H_f(\text{CS}_2) = \Delta H_c(\text{C}) + 2\Delta H_c(\text{S}) - \Delta H_c(\text{CS}_2)$$

$$+88 = -394 + 2(-297) - \Delta H_c(\text{CS}_2)$$

$$\Delta H_c(\text{CS}_2) = \underline{-1076 \text{ kJ mol}^{-1}}$$

- (iii) Carbon disulfide reacts explosively with nitrogen monoxide (NO) according to the following equation



Given the enthalpy change of this reaction is $-1243 \text{ kJ mol}^{-1}$, calculate the enthalpy change of formation of NO.

$$\Delta H_r = \Sigma n \Delta H_f (\text{products}) - \Sigma m \Delta H_f (\text{reactants})$$

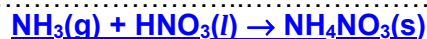
$$-1243 = \Delta H_f(\text{CO}) + 2\Delta H_f(\text{SO}_2) + 5/2(0) - [\Delta H_f(\text{CS}_2) + 5\Delta H_f(\text{NO})]$$

$$= -111 + 2(-297) - [(+88) + 5\Delta H_f(\text{NO})]$$

$$\Delta H_f(\text{NO}) = \underline{+90 \text{ kJ mol}^{-1}}$$

6 Ammonium nitrate is an important fertiliser made by the acid-base reaction between ammonia and nitric acid.

- (a) (i) Write an equation for the production of ammonium nitrate from ammonia and nitric acid.



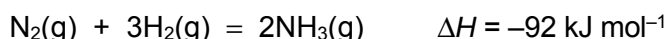
- (ii) Describe the meaning of the terms *Bronsted acid* and *Bronsted base*. Identify the acid and base in the reaction above using Bronsted-Lowry theory.

Bronsted acid is a proton donor and base is a proton acceptor

Acid – HNO₃ Base – NH₃

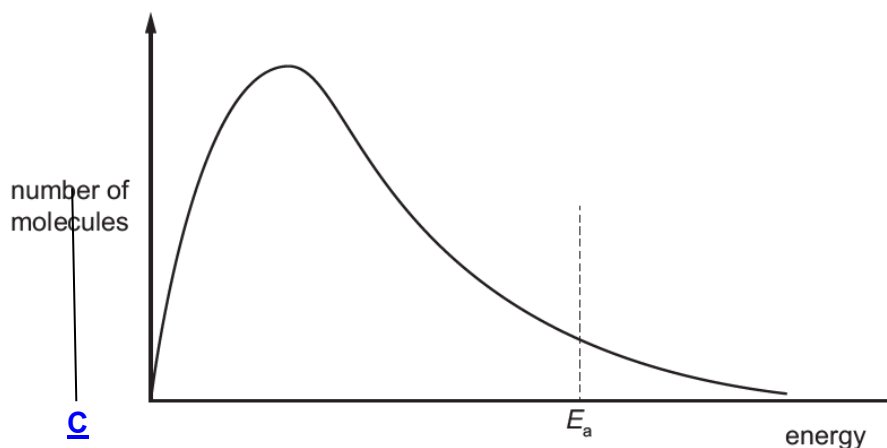
The ammonia for this reaction is produced by the Haber process and the nitric acid is produced by the oxidation of ammonia.

- (b) The Haber process involves a reaction between nitrogen and hydrogen at a temperature of 450 °C and a pressure of 200 atm.



The diagram below shows the Boltzmann distribution curve for the uncatalysed reaction between nitrogen and hydrogen.

energy of the reaction is labelled as E_a .



- (i) The Haber process is an example of a large-scale gaseous reaction that is catalysed. State the catalyst used.

Iron / iron oxide

- (ii) Explain the meaning of the term *activation energy*.

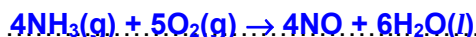
Activation energy (E_a) is the minimum amount of energy that reactants must possess before a reaction can occur.

- (iii) On the energy axis of the graph above, clearly mark the position of the activation energy of hydrogen with the letter **C**.

- 6 (b) (iv) At a higher reaction temperature, the rate of production of ammonia would be greater. Explain why this is the case, referring to the fact that it would increase the rate of production of ammonia.

Production of NH_3 /forward reaction is exothermic
Position of equilibrium will shift left and yield would reduce at higher temperature.

- (c) The first stage in the production of nitric acid involves the reaction of ammonia with oxygen to form nitric oxide. Write an equation for this reaction and use oxidation numbers to show that it is a redox reaction.



The oxidation number of N increases from -3 in NH_3 to +2 in NO .

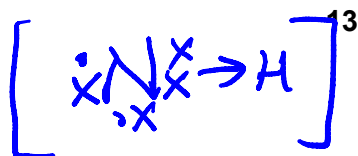
Hence, oxidation has taken place.

The oxidation number of O decreases from 0 in O_2 to -2 in H_2O .

Hence, reduction has taken place.

- (d) (i) Draw a dot-and-cross diagram of the ammonium ion. Show the outer electrons only.





- (ii) State the shape of an ammonium ion and give the H–N–H bond angle.

shape..... **tetrahedral**
 bond angle..... **109/109.5°**

Section B

Answer **one** question from this section, in the space provided.

- 7 (a) Sodium, silicon and chlorine are all elements in Period 3 of the Periodic Table. Describe the variation in atomic radius, melting point, electrical conductivity and bonding of sodium, silicon and chlorine.

For
Examiner's
Use

Atomic radius:

Atomic radius decreases from Na to Si to Cl due to increase in nuclear charge while the shielding effect by inner shell electrons remains almost constant across the period.

Melting point:

Na has a relatively high melting point, Si has the highest melting point, Cl₂ has the lowest melting point

Electrical conductivity:

Na is a good electrical conductor due to presence of mobile electrons in its metallic structure.

Si is a semi-conductor and Cl is a non-electrical conductor due to absence of mobile charge carriers in its structure

Bonding:

Na has strong electrostatic forces of attraction between the delocalised electrons and Na⁺ ions.

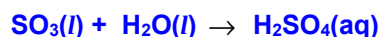
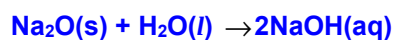
Si has strong covalent bonds between Si atoms.

Cl₂ molecules consists of Cl atoms are bonded by strong covalent bonds, weak induced dipole-induced dipole interaction/attraction exist between Cl₂ molecules.

[5]

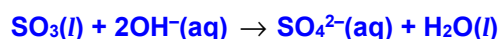
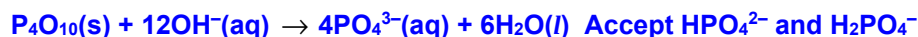
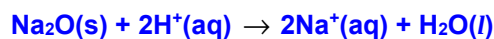
- (b) Three of the oxides of Period 3 elements are sodium oxide, phosphorus(V) oxide and sulfur trioxide.

- (i) Write equations for the reactions of each of these oxides with water.



[3]

- (ii) Write an equation for the reaction of each oxide with either sodium hydroxide or hydrochloric acid.



[3]

- 7 (b) (iii) Describe the variation in behaviour shown by these oxides across Period 3 as shown in (b)(ii).

From basic in Na₂O to acidic in SO₃ and P₄O₁₀

.....

.....

[1]

- (c) (i) Write a balanced equation to represent the lattice energy of sodium chloride.



[1]

- (ii) Explain why the lattice energy of magnesium oxide is much more exothermic than that of sodium chloride.

Ions in MgO are **Mg²⁺** and **O²⁻**

Ions in NaCl are **Na⁺** and **Cl⁻**.

$$|L.E.|\propto \frac{q_+q_-}{r_+ + r_-}$$

charge of Na⁺ is smaller than that of Mg²⁺

charge of Cl⁻ is smaller than that of O²⁻

radius of Na⁺ is greater than that of Mg²⁺.

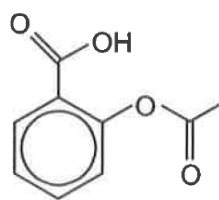
radius of Cl⁻ is greater than that of O²⁻.

Therefore, the lattice energy of MgO is expected to be more exothermic.

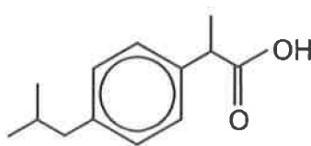
[2]

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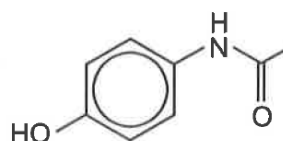
- (d) The pharmaceuticals aspirin, ibuprofen and paracetamol are all painkillers.



aspirin



ibuprofen



paracetamol

Infra-red absorptions can be used to identify functional groups in organic compounds. For example, ethyl ethanoate shows absorptions at $1050\text{--}1330\text{ cm}^{-1}$ and $1710\text{--}1750\text{ cm}^{-1}$ as shown in the *Data Booklet*.

- (i) Identify an infra-red absorption range that will be shown by aspirin and ibuprofen but **not** by paracetamol using the *Data Booklet*.

1680 – 1730 / 2500 – 3000 / 1210 – 1440 (cm^{-1})

[1]

- (ii) Identify two of the infra-red absorption range that will be shown by paracetamol but **not** by aspirin or ibuprofen using the *Data Booklet*.

Any 2 from:

3300 – 3500 / 1640 – 1690 / 3200 – 3600 / 3580 – 3650 (cm^{-1})

[1]

- 7 (d) (iii) Identify a type of reaction which will occur with both paracetamol and aspirin but **not** with ibuprofen.

Hydrolysis

[1]

- (iv) Write an equation for the reaction of ethanol with ethanoic acid. State the conditions required.



Conditions: conc H_2SO_4 , heat under reflux

[2]

[Total: 20]

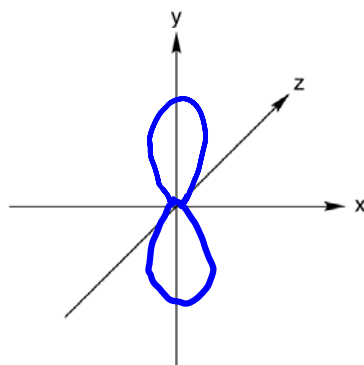
8 This question is about phosphorus and its compounds.

- (a) (i) State the full electronic configuration of phosphorus.

$1s^2 2s^2 2p^6 3s^2 3p^3$

[1]

- (ii) On the axes below, sketch the shape of the singly-occupied orbital in an atom of phosphorus at ground state.



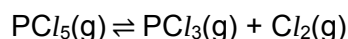
[1]

- (iii) The first ionisation energy of phosphorus is higher than that of sulfur. Explain why.

Mutual repulsion between the paired 3p electrons in sulfur makes it easier to remove the one of the paired 3p electrons from sulfur than to remove the unpaired 3p electron from phosphorus which does not experience such repulsion......

[1]

- (b) The dissociation of phosphorus(V) chloride into phosphorus(III) chloride and chlorine gas is reversible:

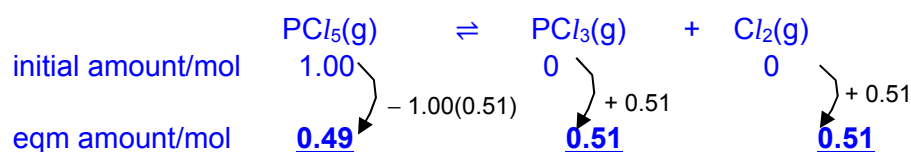


The reaction is an example of a *dynamic equilibrium*.

- (i) Explain the meaning of the term *dynamic equilibrium*.

Rate of forward reaction equals to rate of backward reaction [1]

- (ii) In an experiment, 1.00 mol of PCl_5 vapour was heated in a closed 5.00 dm³ flask at 500 K until equilibrium had been established. When equilibrium is established, the percentage conversion of PCl_5 is found to be 51 %. Calculate the equilibrium constant, K_c , at this temperature and state its units.



$$K_c = \frac{[\text{PCl}_3][\text{Cl}_2]}{[\text{PCl}_5]} = \frac{\left(\frac{0.51}{5}\right)\left(\frac{0.51}{5}\right)}{\left(\frac{0.49}{5}\right)} = \underline{\underline{0.106 \text{ mol dm}^{-3}}}$$

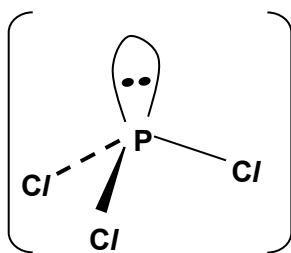
[2]

- 8 (b) (iii) State and explain what would happen to the equilibrium position if the initial concentration of chlorine was increased.

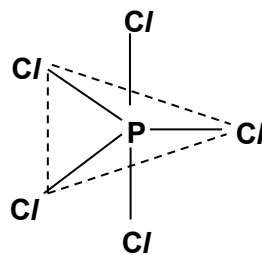
Equilibrium position shift to the right to remove some of the Cl_2

..... [1]

- (iv) Draw and name the shapes of PCl_3 and PCl_5 .



trigonal pyramidal



trigonal bipyramidal

[2]

- (v) Explain why PCl_5 sublimes at 160°C while PCl_3 boils at a lower temperature of 76.1°C.

Both PCl_5 and PCl_3 have simple molecular structures.

Due to the larger number of electrons in the PCl_5 molecule/ larger electron cloud of PCl_5 molecule..... [1]

more energy/ larger amount of energy is required to overcome/break the stronger induced dipole-induced dipole interaction/attraction between PCl_5 molecules than the weaker permanent dipole-permanent dipole interaction/attraction between PCl_3 molecules

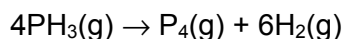
- (d) Phosphorus compounds are used in the pharmaceutical industries. For example, solutions containing hydrogen phosphates are used as buffers found in the hydrophilic gel used in skin patches. Write equations to show how the solution containing $\text{H}_2\text{PO}_4^- / \text{HPO}_4^{2-}$ can act as a buffer when H^+ and OH^- ions are added respectively to the solution.

On addition of H^+ : $\text{HPO}_4^{2-} + \text{H}^+ \rightarrow \text{H}_2\text{PO}_4^-$

On addition of OH^- : $\text{H}_2\text{PO}_4^- + \text{OH}^- \rightarrow \text{HPO}_4^{2-} + \text{H}_2\text{O}$

[2]

- 8 (e) Phosphine, PH_3 , is a widely used fumigant. The thermal decomposition of phosphine into phosphorus and hydrogen is a *first-order reaction*.



- (i) Explain the term in *italics*.

A first-order reaction refers to one in which the order of reaction with respect to the reactant is one in the rate equation.

OR

In a first-order reaction, the power to which a concentration of reactant is raised in the rate equation is one.

OR

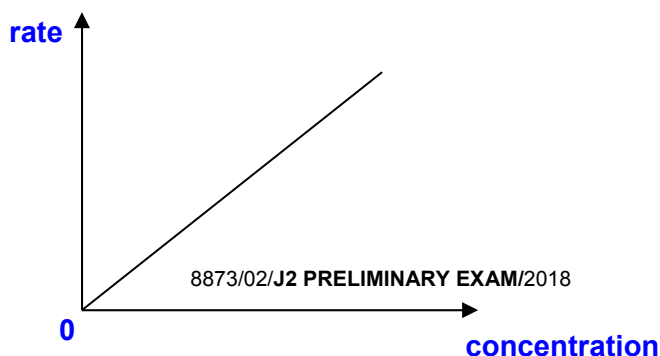
Consider an experimentally determined rate equation,

$$\text{Rate} = k[\text{A}]^1,$$

for a first-order reaction, order of reaction with respect to A = 1.

[1]

- (ii) Sketch a graph of rate against concentration for the decomposition reaction.



[1]

The half-life of the reaction is 35.0 seconds at 680 °C. Calculate

- (iii) the rate constant for the reaction.

$$k = \ln 2 / t_{1/2} = \ln 2 / 35.0 = \underline{0.0198 \text{ s}^{-1}} \quad [1]$$

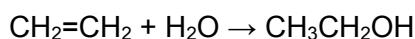
- (iv) the time taken for 87.5% of the phosphine to decompose.

$$100\% \rightarrow 50\% \rightarrow 25\% \rightarrow 12.5\%$$

$$\text{Time taken} = 4 \times 35.0 = \underline{140 \text{ s}} \quad [1]$$

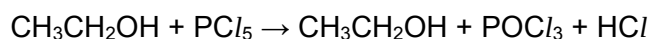
- (f) Phosphorus compounds are also used in organic reactions, as shown in the following reactions. Suggest the type of reaction occurring in each case.

- (i) Phosphoric(V) acid, H_3PO_4 , is used as a catalyst in the reaction.



addition

- (ii) Phosphorus(V) chloride reacts with ethanol.



substitution

- (g) Describe, in terms of orbital overlap, the bonding of the two carbon atoms of the C=C bond in $\text{CH}_2=\text{CH}_2$. A clearly labelled diagram may clarify your answer.

In ethene, one sp^2 orbital of C atom overlaps head-on with the sp^2 orbital of another C atom, forming a C–C sigma bond.

The unhybridised 2p orbital of C atom overlaps side-way with the unhybridised 2p orbital of another C atom, forming a C–C π bond.

Or accept clearly labelled diagram

[2]

[Total: 20]

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